

LSTMs, GRUs, and Gating Mechanisms

Scaler School of Technology | Deep Learning I | Week 6 — Session 1/2 | Post-Lecture Review & Practice



Key Takeaways

1 LSTM: Controlled Memory via Gates

The cell state provides a gradient highway (additive updates only). Three sigmoid gates—forget (f_t), input (i_t), output (o_t)—control what to erase, write, and expose. Key equation: $C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$.

2 GRU: Same Idea, Fewer Parts

Two gates (reset r_t , update z_t) with no separate cell state. The update gate handles both forgetting and adding in one operation: $h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t$. 75% of LSTM parameters, similar performance.

3 Architecture Choices

Bidirectional = full context from both directions (only when whole sequence is available). Deep/stacked = hierarchical features (2–4 layers typical). The 2015–2017 NLP standard: 2-layer bidirectional LSTM with dropout.

Test Your Understanding

Try answering these without looking at the slides. Answers are on page 3.

- Q1.** Write the cell state update equation for LSTM. Why is the additive form important for gradient flow?
- Q2.** What happens if the forget gate outputs all 1s for every timestep? What about all 0s?
- Q3.** A GRU has an update gate value $z_t = 0$ at some timestep. What is h_t in that case, and what does this mean for gradient flow?
- Q4.** You're building a real-time speech-to-text system. Should you use a bidirectional LSTM? Why or why not?
- Q5.** You have a text classification task with short reviews (10–20 words). Would you use a vanilla RNN, GRU, or LSTM? Justify your choice.

Work through these in a Colab notebook. No starter code — figure out the approach from what you learned in the lecture.

Exercise 1: LSTM vs RNN on Sequence Length

Generate the “remember the first token” task (signal at position 0, noise for T steps, predict signal at end). Train both a vanilla RNN and an LSTM for $T \in \{20, 50, 100, 200\}$. Plot test accuracy vs. sequence length for both. **Expected output:** A line chart showing RNN accuracy dropping to 50% while LSTM stays high until much longer sequences.

Exercise 2: Visualize Forget Gate Behavior

Train an LSTM on a character-level language model (use any short text, e.g., Shakespeare). After training, feed a sentence through the model and extract forget gate activations at each timestep. Plot as a heatmap. **Expected output:** A heatmap showing the forget gate responding to sentence boundaries and context switches.

Exercise 3: Bidirectional Advantage

Create a sequence tagging task where future context helps (e.g., predicting whether each token is inside a bracket pair “[]”). Compare unidirectional LSTM vs. bidirectional LSTM accuracy. **Expected output:** Bidirectional should achieve significantly higher accuracy since knowing where brackets close requires future context.

Challenge Problem

Sentiment Analysis with Architecture Comparison

Build a complete sentiment classifier on IMDB reviews (or a similar dataset) and compare architectures:

1. Download IMDB dataset (use torchtext or HuggingFace datasets)
2. Implement an embedding layer + recurrent model + FC classifier
3. Train and evaluate 4 variants: (a) single-layer GRU, (b) single-layer LSTM, (c) 2-layer bidirectional LSTM, (d) 2-layer bidirectional GRU
4. Report: accuracy, training time per epoch, and total parameters for each
5. Plot training curves (loss vs. epoch) for all 4 on the same chart

Bonus: Add gradient clipping and dropout. Measure how much each improves the best model.

- **LSTM Original:** Hochreiter & Schmidhuber (1997), “Long Short-Term Memory” — The foundational paper. Dense but worth reading Section 1–3 for motivation.
- **GRU Original:** Cho et al. (2014), “Learning Phrase Representations using RNN Encoder-Decoder” — Introduces GRU as part of the encoder-decoder framework.
- **Empirical Comparison:** Chung et al. (2014), “Empirical Evaluation of Gated Recurrent Neural Networks on Sequence Modeling” — Head-to-head LSTM vs. GRU on multiple tasks.
- **Visual Guide:** Colah (2015), “Understanding LSTM Networks” — The best visual explanation of LSTM gates. Highly recommended.

Online Resources & Tools

- **PyTorch nn.LSTM docs** — API reference with shape conventions
- **Colah’s LSTM blog** — Best visual guide to gates
- **D2L.ai Ch. 10** — Interactive LSTM/GRU chapter with code
- **Jay Alammar’s Illustrated Guide** — Step-by-step visual walkthroughs
- **Karpathy’s char-rnn** — Classic LSTM demo with code
- **distill.pub** — Research-quality interactive visualizations

Fill the Gaps — Prerequisite Refreshers

Felt lost on a specific concept? These resources will help you catch up.

- **Confused about why sigmoid acts as a gate?** Sigmoid maps any input to $[0, 1]$. Multiplying by 0 = block, by 1 = pass through. Review activation functions from Week 1 Session 2.
- **Fuzzy on element-wise operations ($\odot$)?** This is just multiplication/addition applied independently to each dimension. No matrix multiply — each element of f_t scales the corresponding element of C_{t-1} .
- **Vanishing gradient still unclear?** Revisit last session’s (Week 5 Session 2) material on BPTT and the repeated multiplication problem. The LSTM cell state avoids this by using addition instead.
- **Not sure about concatenation $[h_{t-1}, x_t]$?** This just stacks the two vectors into one longer vector before passing through a weight matrix. If h is 128-dim and x is 50-dim, the concatenation is 178-dim.

Answers — Test Your Understanding

- A1.** $C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$. The additive form means $\partial C_t / \partial C_{t-1} = f_t$ (element-wise), avoiding the repeated matrix multiplication that causes vanishing gradients in vanilla RNNs.
- A2.** All 1s: cell state is preserved perfectly ($C_t = C_{t-1} + \text{new}$) — the LSTM never forgets. All 0s: cell state is wiped every step ($C_t = i_t \odot \tilde{C}_t$) — it behaves like a memoryless network.
- A3.** When $z_t = 0$: $h_t = (1 - 0) \odot h_{t-1} + 0 \odot \tilde{h}_t = h_{t-1}$. The state passes through unchanged — perfect gradient flow (identity mapping).
- A4.** No — real-time speech processing is streaming (you don’t have the full audio upfront). Bidirectional requires the complete sequence. Use a unidirectional LSTM/GRU instead.
- A5.** For short reviews (10–20 words), a GRU is likely sufficient. The sequences are short enough that even vanilla RNN might work, but GRU provides safety margin with minimal overhead. LSTM’s extra complexity is unlikely to help for such short sequences.